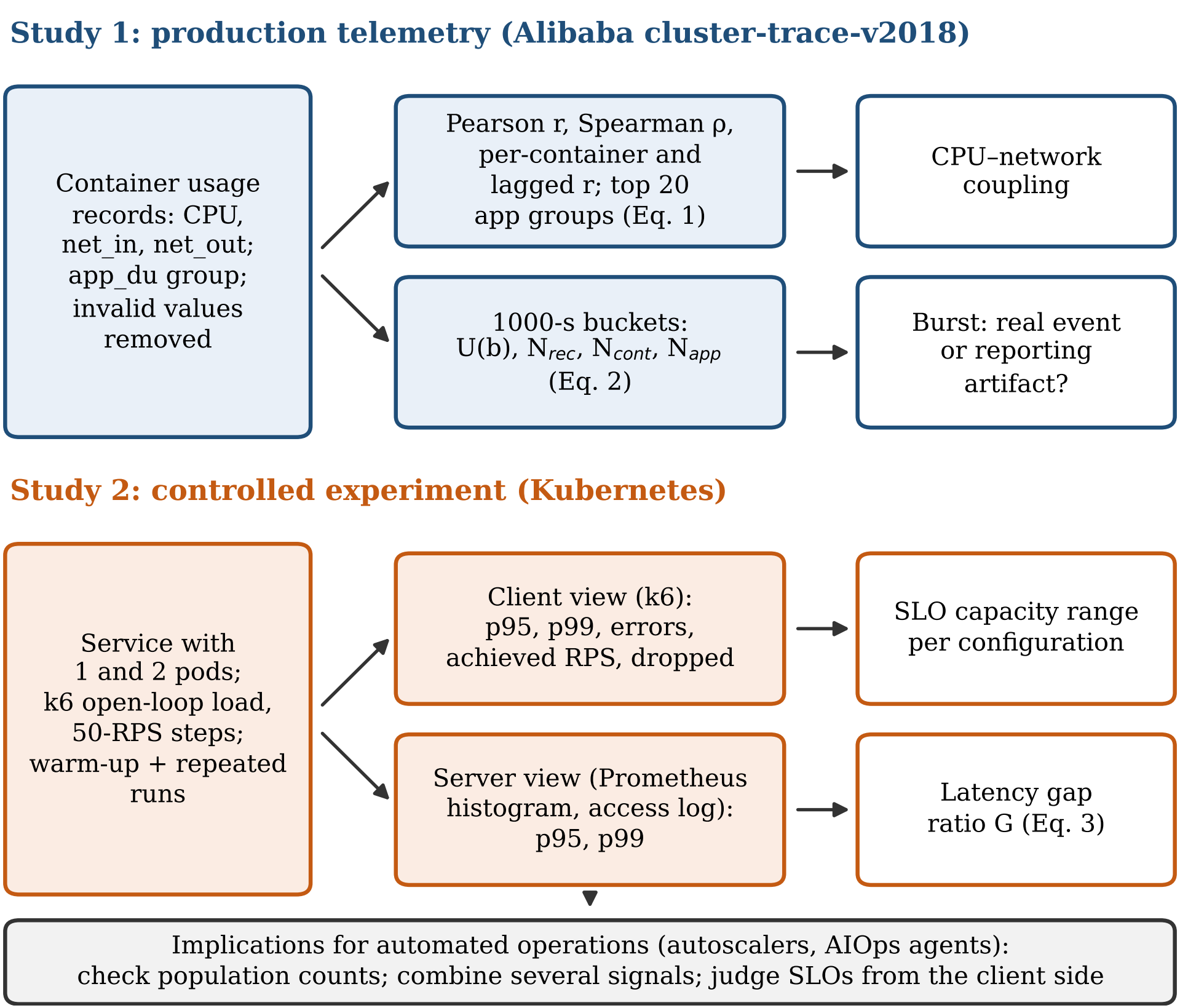
Study 1: production telemetry (Alibaba cluster-trace-v2018)
Container usage records: CPU, net_in, net_out; app_du group; invalid values removed
Pearson r, Spearman ρ, per-container and lagged r; top 20 app groups (Eq. 1)
CPU–network coupling
1000-s buckets: U(b), $N_{rec}$, $N_{cont}$, $N_{app}$ (Eq. 2)
Burst: real event or reporting artifact?
Study 2: controlled experiment (Kubernetes)
Service with 1 and 2 pods; k6 open-loop load, 50-RPS steps; warm-up + repeated runs
Client view (k6): p95, p99, errors, achieved RPS, dropped
SLO capacity range per configuration
Server view (Prometheus histogram, access log): p95, p99
Latency gap ratio G (Eq. 3)
Implications for automated operations (autoscalers, AIOps agents): check population counts; combine several signals; judge SLOs from the client side